

SHARIF UNIVERSITY OF TECHNOLOGY

Fault-Tolerant Deep Neural Network Hardware Accelerator Design

for Image and Video Processing Applications

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Three Critical Walls → One Revolutionary Solution

! Moore's Wall

PHYSICAL LIMIT

Transistor scaling hit atomic barriers. Gate oxide only a few atoms thick — leakage becomes prohibitive below 7 nm.

! Von Neumann Wall

BUS & MEMORY BOTTLENECK

Memory and processor are physically separated. Data shuttling consumes up to 200× more energy than the computation itself.

! Energy Wall

POWER LIMIT

180+ ZB of data by 2025. Edge AI needs real-time inference at µW budgets — impossible with traditional architectures.

⏳ *PARADIGM SHIFT*



THE SOLUTION

Neuromorphic Computing Paradigm

Spiking Neural Networks — the third generation of artificial intelligence

⚡Event-Driven • 🧠 Brain-Inspired • 🔋 µW Power

Sub-10 nm Silicon Meets Mission-Critical Edge AI

Permanent fault sources in nanoscale VLSI

Manufacturing Defects

Lithography errors and process variation at sub-7 nm

Aging — NBTI / HCI

Threshold shift and increased leakage over years of use

Voltage & Temperature

IR drops, hotspots and timing violations

Radiation — SEU / SET

Alpha particles and neutron strikes cause bit-flips

Where failure equals catastrophe

A single stuck-at fault can collapse an entire system

Aerospace

Cosmic radiation — no repair is possible in orbit

Medical Implants

Neuroprosthetics — replacement means invasive surgery

Autonomous Vehicles

Real-time safety — a failure can cost lives

Industrial IoT / Edge

Unattended operation in extreme environments

Spiking Neural Networks — The Third Generation

Feature	Traditional ANN	Spiking NN (SNN)
Information coding	Continuous real values	Discrete spikes (temporal)
Computation	MAC — multiply-accumulate	Accumulate — event-driven
Power consumption	Watts — always active	µW — sparse, event-driven
Biological plausibility	Low — rate abstraction	High — spike-timing dynamics
Hardware cost	DSP slices required	Shift-add only

⚡ Event-Driven

Neurons compute only when a spike arrives — idle power is near zero

🧠 Biological Dynamics

20+ firing patterns (RS, IB, CH, FS) far beyond simple rate coding

🔋 Ultra-Low Power

µW to mW range — up to 1000× more efficient than equivalent ANNs

🏭 Temporal Coding

Information lives in precise spike timing: phase, latency, synchrony

Five Critical Research Gaps

1	**Biological Fidelity vs. Hardware Reliability**	**CURRENT LIMITATION** SNN fault-tolerance research focuses almost exclusively on simplistic LIF neurons.	**THE GAP** No study addresses the fault vulnerability of nonlinear, biologically plausible dynamics needed for neuroprosthetics and BCIs.
2	**Lack of Physically Accurate Reliability Assessment**	**CURRENT LIMITATION** Tools rely on high-level software simulation or simple binary (0,1) logic.	**THE GAP** No framework models bridging and stuck-open faults with full 4-valued logic (0,1,X,Z) at gate level.
3	**Classical FT Clashes with Edge / Implant Constraints**	**CURRENT LIMITATION** TMR imposes more than 200% area and power overhead.	**THE GAP** Prohibitive for µW-scale edge devices, neuroprosthetics and autonomous systems.
4	**Absence of Predictable, Intrinsic Self-Healing**	**CURRENT LIMITATION** Mitigation relies on heuristic retraining, offline mapping or external redundancy.	**THE GAP** No intrinsic, mathematically guaranteed mechanism reacts the instant a hardware fault occurs.
5	**Disconnect: Neuron Resilience → System Execution**	**CURRENT LIMITATION** Fault tolerance is isolated at the software algorithm level or the hardware module level — never both.	**THE GAP** No unified architecture bridges neuron-level biological resilience with system-level physical implementation.

Five Research Objectives

Hardware-Aware Design of Biologically Rich Neurons

GAP 1

A hardware-optimised, multiplier-less architecture for highly nonlinear, biologically plausible spiking models (Izhikevich) that keeps biological fidelity without sacrificing silicon efficiency.

Mathematically Guaranteed Intrinsic Self-Healing

GAP 4

An autonomous homeostatic regulator with formal Lyapunov stability proofs, ensuring neuron firing stays bounded and stable even under severe permanent hardware faults.

Rigorous, Physically Accurate Fault Evaluation

GAP 2

A multi-valued (0,1,X,Z) gate-level fault-injection framework (GLFI) able to simulate realistic physical defects — bridging and stuck-open — at netlist level.

Ultra-Lightweight Fault Tolerance for Edge Constraints

GAP 3

Mission-critical reliability at a fraction of the area and power overhead of classical redundancy (TMR), making it viable for µW-scale edge AI and neuroprosthetics.

Cross-Layer System Integration for Real-Time Inference

GAP 5

Self-healing homeostatic neuron cores integrated into a unified, time-multiplexed SNN accelerator — bridging device-level resilience to system-level execution.

§ 2 · LITERATURE REVIEW

Methodology Roadmap

Three prior works and the proposed framework

1

SPANNER

Astrocyte self-repair
on FPGA hardware

2

SoftSNN

Transient soft-error mitigation
in SNN accelerators

3

Spyrou

Neuron fault tolerance
via dropout and TMR

4

Our Method

H-ASIZH + GLFI Engine
with FAASS integration

Table 1 — Fault Tolerance: Critical Comparative Analysis

Reference	FT Method	Reliability & Key Results	PPA Report	Platform	Critical Weakness
SPANNER [3] (Liu 2018)	Astrocyte self-repair (e-SP + DSE, Li-Rinzel ODE)	40% fault tolerance 80% fault → 20% perf. drop Graceful degradation	3552 slices 45 DSPs (125% overhead) Claims 15.4% — misleading	FPGA LIF neuron	🚫 Baseline not optimised → relative % deceptive 🚫 15 ODEs = heavy compute
Spyrou [5] (2021)	Fault hopping + dropout + TMR (output layer only)	40% dead neurons tolerated N-MNIST: no accuracy loss	No PPA report Analog schematic only 1 transistor / neuron claim	Hybrid (Analog + PyTorch)	🚫 Not real-time — offline test 🚫 Saturated → dead = data loss 🚫 No digital PPA validation
SoftSNN [4] (Putra 2022)	Bound-and-Protect (BnP) Weight clipping + reset disable	< 3% accuracy drop at 10% soft-error rate	14–18% area 1.6× energy Relative only — no absolutes	ASIC 65 nm LIF neuron	🚫 Only transient soft errors 🚫 No fault-injection details 🚫 No digital PPA validation
RescueSNN [15] (Putra 2023)	FAM + bit-shuffling Fault-aware mapping	80% accuracy recovery < 25% throughput loss at 50% permanent fault rate	36.5% area (36.2% = ECU regs) 30% energy overhead HEB blocks: 0.3% area	ASIC 65 nm LIF neuron	🚫 Requires offline fault map 🚫 No on-chip learning 🚫 ECU dominates area cost
Naghieh (2023)	BoF astrocyte-based working memory + self-repair	20% working-memory gain 60% synaptic fault tolerance	Not reported Theoretical only	Simulation (software)	🚫 No hardware implementation 🚫 No PPA data 🚫 Purely theoretical
Spyrou [20] (2022)	Bit-flip injection + BER analysis	Critical regions identified: splitter, router, kernel	Selective protection enabled by the analysis	FPGA HW accelerator	🚫 Analysis only — no FT solution 🚫 Memory-only fault model 🚫 No combinational coverage
This Work (H-ASIZH)	**Homeostasis + Restoring Organ**	**95.8% survival (GLFI) 84.3% on-chip learning Lyapunov stability proof**	**4093 cells (absolute) 91.4 mW at 62 MHz SkyWater 130 nm**	**ASIC 130 nm Izhikevich neuron**	✅ Real-time dynamic FT ✅ Mathematical proof ✅ Exhaustive GLFI · absolute PPA

Table 2 — Fault Injection & Analysis Frameworks

Tool / Reference	Injection Level	Open Source	4-Valued Logic	VHDL & Verilog	Key Limitation / Primary Innovation
SpikeFI (Spyrou 2024) [20]	Software (PyTorch)	●	○	○	Lacks physical hardware fidelity — software-level injection only
RADSAFiE (Monopoli 2025) [10]	Netlist (FPGA)	○	○	○	Vendor-locked to Vivado; cannot model bridging faults
SHADOWFI (Guerrero 2025) [11]	Netlist (ASIC)	●	○	○	Verilog only; fails to capture X / Z physical states
El-Sayed [21,23] (2019-2023)	Netlist / transistor	●	○	○	Lacks an automated GLFI campaign architecture
Spyrou [20] (2022)	Memory (FPGA)	○	○	○	Restricted to memory SEUs; no combinational fault injection
This Work GLFI Engine	**Gate-level simulation**	●	●	●	**Multi-agent LLM testbench generation · Shadow-Hook FI engine (saboteur weaving) 4-valued logic (0,1,X,Z) for bridging and stuck-open · VHDL + Verilog + Cocotb Online FAASS service: faass.viracoai.net**

● supported ○ not supported

§ 3 · METHODOLOGY

A Two-Part Methodology

Reference works are dissected first, then the proposed architecture is built on what they leave open.

A Reference Papers

SPANNER · SoftSNN · Spyrou

One slide per paper: key figures, five takeaways, and the weakness that motivates the next step.

B Proposed Methodology

H-ASIZH · GLFI Engine

Affine hardware mapping, homeostatic control with a Lyapunov proof, and an exhaustive GLFI campaign.

SPANNER [3]: Astrocyte-Mediated Self-Repair (Liu 2018)

1. **The astrocyte–neuron tripartite synapse mimics the brain's adaptive homeostasis for fine-grained fault tolera**
2. **When synapses fail, the astrocyte sends indirect feedback (e-SP) that raises the release probability of the hea**
3. **Fully modular digital hardware — neuron, synapse, astrocyte — with pipelining that boosts throughput by 49**
4. **Graceful degradation: even at 80% synaptic fault density the performance loss is only 20%.**

BUT: 125% DSP overhead, no mathematical stability proof, and all PPA figures are reported relative to an unoptimised baseline.

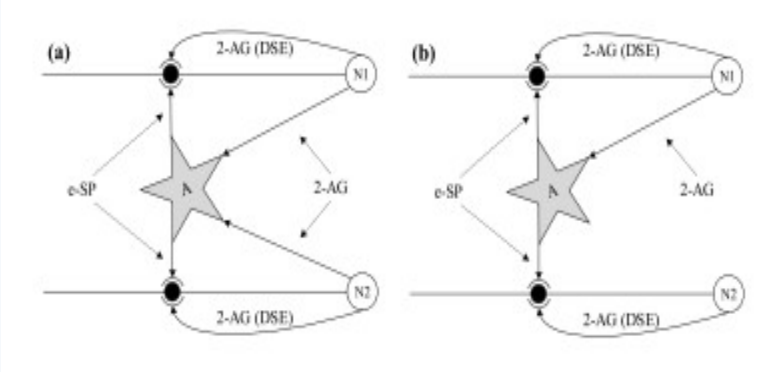


Fig1:Tripartite synapse — self-repair

Fault density	Platform	Average output frequency	
		Neuron 1	Neuron 2
0%	Simulation	7.19	7.20
	Hardware	**7.28**	**7.27**
40%	Simulation	7.38	6.81
	Hardware	**7.37**	**6.88**
80%	Simulation	7.38	5.68
	Hardware	**7.37**	**5.75**

TABLE I — output frequency

	Astrocyte	Synapse & Neuron	DSE generator
Slice LUTs	11394	9865 [12.8%]*	4353
Slice Registers	11666	10383 [17.4%]	4688
Slice	3552	3120 [15.4%]	1394
DSPs	42	45 [125%]	14

* The percentage denotes how many additional resources are required for the synapse & neuron component to implement the self-repairing mechanism.

TABLE II — hardware utilisation

SoftSNN [4]: Bound-and-Protect for Soft Errors (Putra 2022)

1. A run-time methodology that neutralises soft errors (bit-flips) in the weight registers without re-executi
2. Bound-and-Protect clips abnormal weights and disables the neuron output when the critical Vmem rese
3. Radiation-hardened registers and lightweight MUX gates are integrated into the baseline compute engi
4. Accuracy loss stays below 3% at the highest fault rates, with 3× lower latency and 2.3× energy savings.

BUT: only transient soft errors are covered, every PPA figure is a relative percentage, and no exhaustive fault scenario is

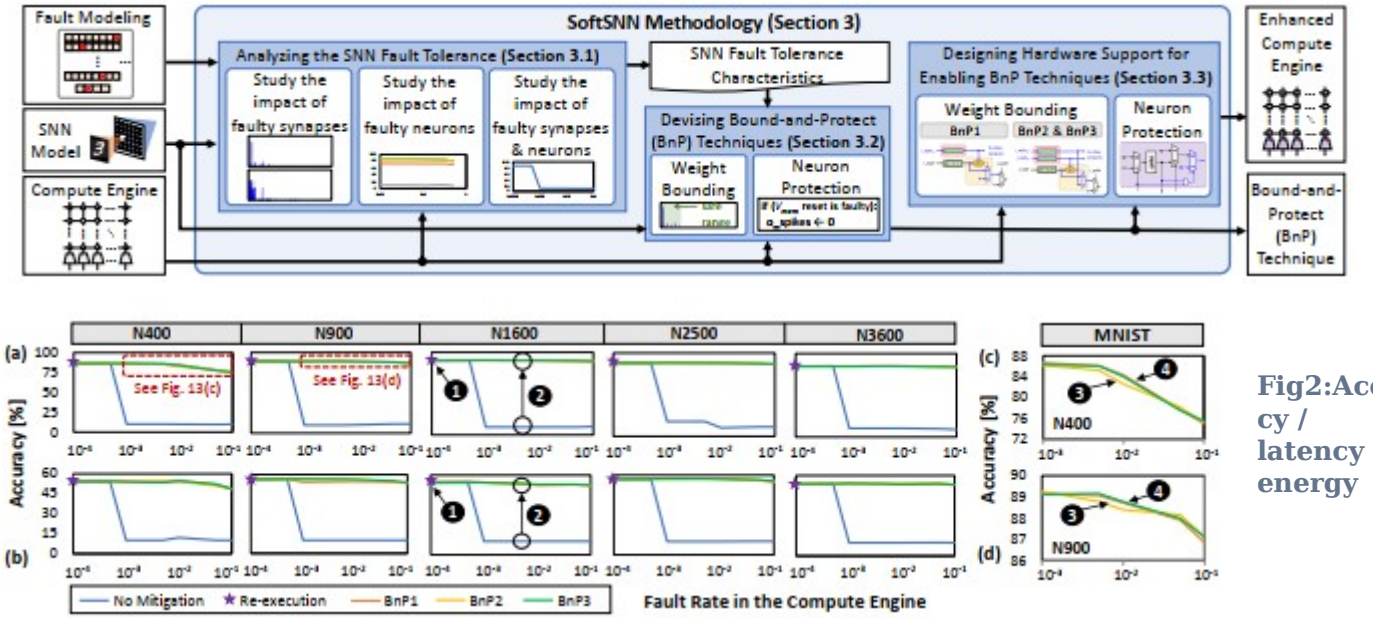


Fig1:Bound-and-Protect flow

Fig2:Accuracy / latency / energy

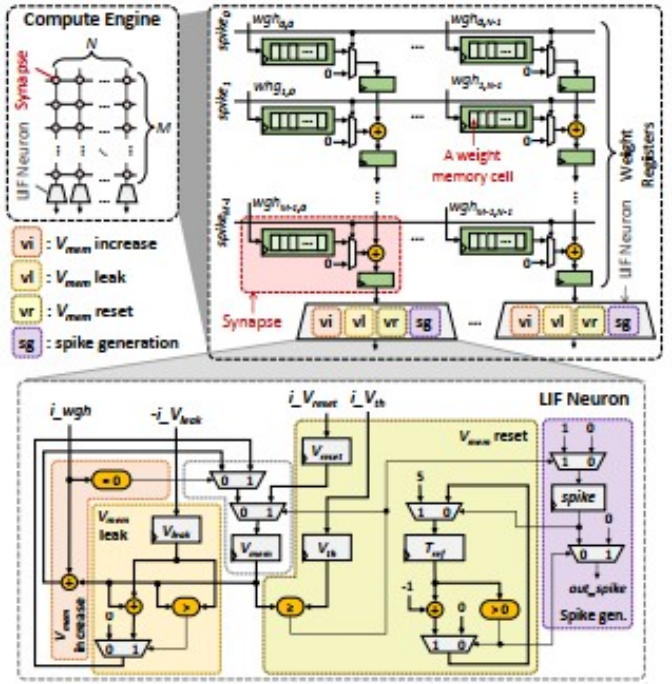
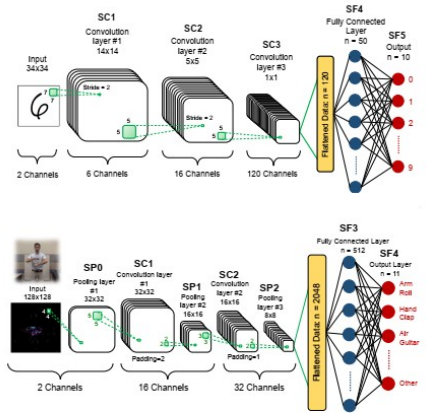


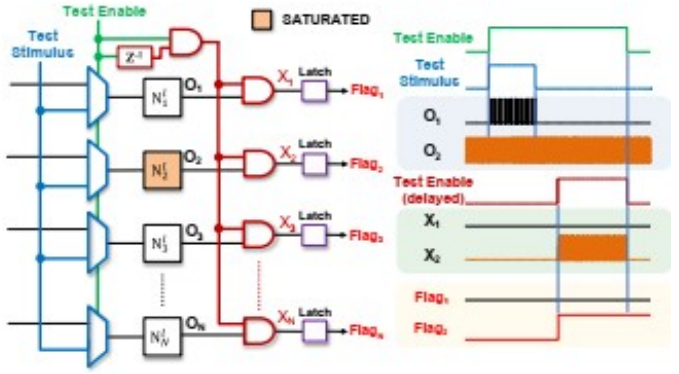
Fig3:SNN accelerator architecture

Spyrou [5]: Neuron Fault Tolerance in SNNs (Spyrou 2021)

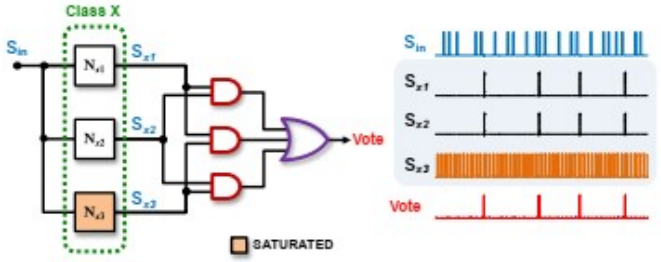
Fault models and dropout effect



I&F neuron + self-test + TMR



Multiple fault scenario results



1. Dual strategy for dead, saturated and timing-variation faults at behavioural level — no hardware redundancy in hidden layers.
2. Fault hopping: dropout training neutralises dead neurons, while online self-test converts saturated neurons into dead ones.
3. A 3-bit counter per neuron performs online saturation detection; TMR is applied only to the output layer (< 1% of neurons).
4. Full accuracy is preserved with 40% of hidden neurons dead — at the cost of just one control transistor per neuron.

BUT: no PPA report, no gate-level fault injection, and behavioural-level validation only — never real silicon.

§ 3-4 · PART B: PROPOSED METHOD

H-ASIZH + GLFI Engine

Homeostatic-Adaptive Spiking Izhikevich Hardware, verified by a gate-level fault-injection engine.

1 Izhikevich Neuron

Nonlinear ODE dynamics
Q7.19 fixed point, no multipliers

2 Homeostasis

Plant–controller structure
Lyapunov UUB proof

3 GLFI Engine

Multi-agent LLM testbenches
Shadow-Hook fault injection

Mathematical Framework — From Biology to Hardware

STEP 1 · Continuous Izhikevich model (biological)

$$\frac{dv}{dt} = 0.04\,v^2 + 5v + 140 - u + I$$

$$\frac{du}{dt} = a(bv - u)$$

$$v \geq 30\ \text{mV} \Rightarrow v \leftarrow c,\quad u \leftarrow u + d$$

Signed range [−70, +30] mV · requires true multipliers

↓ AFFINE SHIFT + SCALE

STEP 2 · Hardware-mapped domain

$$V_{hw} = 64\,(v_{bio} + 65)$$

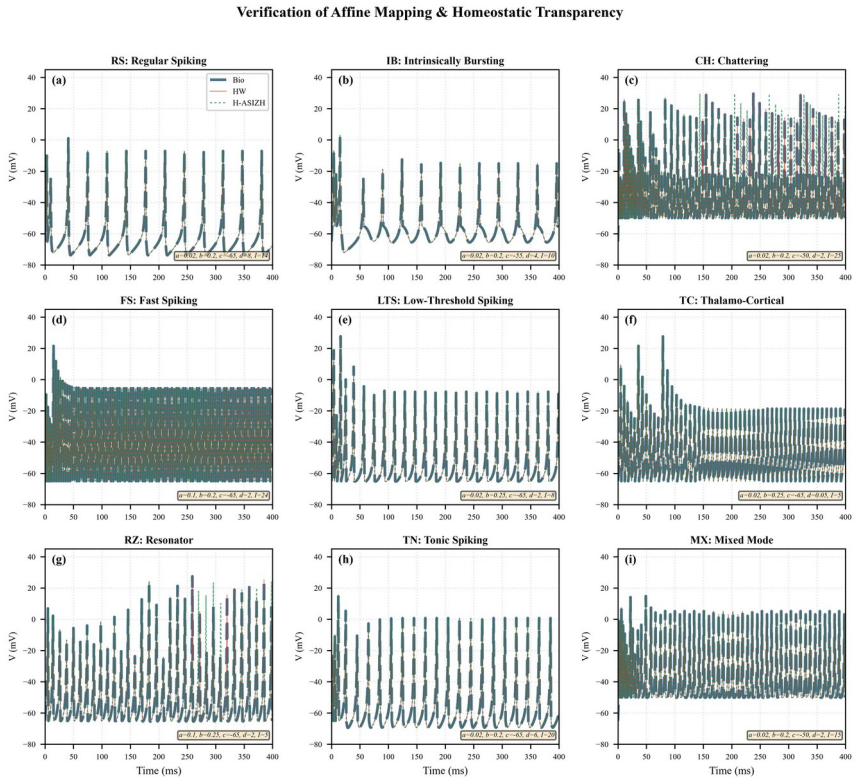
$$\frac{dV_{hw}}{dt} = \frac{1}{64}V_{hw}^2 - 0.2\,V_{hw} + I_{hw} - U_{hw} + \Psi_{bias}$$

$$\frac{dU_{hw}}{dt} = a\,(b\,V_{hw} - U_{hw} - 4160\,b)$$

[−65, +30] mV → [0, 1.48] · all positive, no sign bit

No signed arithmetic · shift-add only · phase portrait preserved

Nine Izhikevich spiking patterns — exact match



All nine patterns reproduced — RMSE < 10^{-13}

The H-ASIZH Homeostatic Controller

$$\alpha = 2^{-6}$$

Leak Rate

Analogue of Ca^{2+} diffusion. Vth decays back to its baseline during normal firing.

$$\gamma = 2^{-3}$$

Penalty Jump

Every spike penalises Vth by γ, so consecutive spikes choke a runaway seizure.

$$\beta = 2^{3} = 8$$

Beta-Brake

Regulates Uhw reset to prevent deep hyperpolarization and avoid dead neurons during faults.

The complete H-ASIZH ODE set

Plant (fast dynamics)

$$\frac{dV_{hw}}{dt} = \frac{1}{64} V_{hw}^2 - 0.2\, V_{hw} + I_{hw} - U_{hw} + \Psi_{bias} + I_{fault}$$

$$\frac{dU_{hw}}{dt} = a\,(b\, V_{hw} - U_{hw} - 4160\, b)$$

Controller (slow dynamics)

$$\frac{dV_{th}}{dt} = \alpha\,(V_{th0} - V_{th}) + \gamma \sum_k \delta(t - t_k)$$

$$V_{hw} \geq V_{th} \Rightarrow V_{hw} \leftarrow c_{hw},$$

$$U_{hw} \leftarrow U_{hw} + d_{hw} + \beta\,(V_{th} - V_{th0})$$

Lyapunov Stability Proof

STEP 1 · Averaging theorem

$$\frac{dV_{th}}{dt} = \alpha(V_{th0} - V_{th}) + \gamma\, r(t)$$

Dirac pulses are mapped onto a continuous mean firing rate r(t).

∂r/∂Vth < 0 — raising the threshold always reduces firing.

STEP 2 · Lyapunov function

$$\mathcal{V}(V_{th}) = \frac{1}{2}(V_{th} - V_{th}^{*})^2$$

Equilibrium: α(Vth0 − Vth*) + γ r* = 0

Positive definite for Vth ≠ Vth*, and zero at equilibrium.

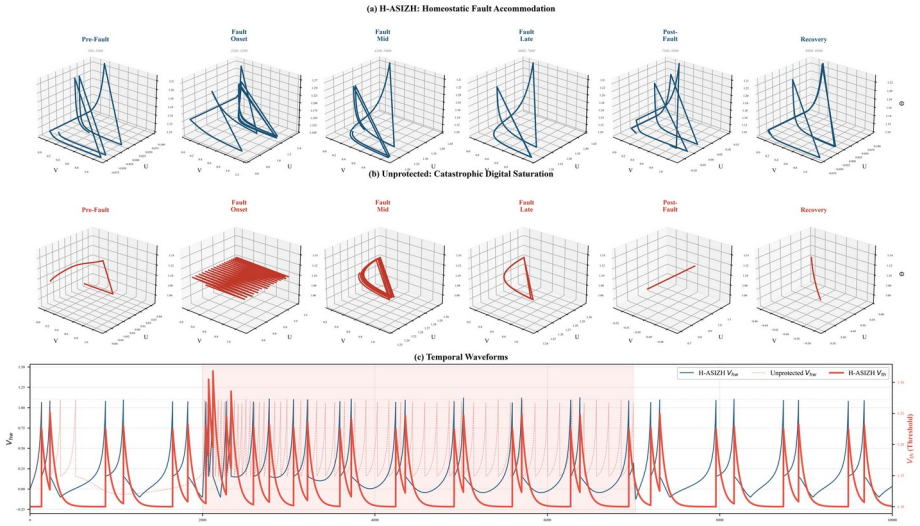
STEP 3 · Lie derivative

$$\dot{\mathcal{V}} = (V_{th} - V_{th}^{*})\,[\,-\alpha(V_{th} - V_{th}^{*}) + \gamma(r - r^{*})\,]$$

The derivative is negative in both hyperactive and hypoactive states.

Q.E.D. — UUB guaranteed, digital saturation is impossible.

2-D limit cycle → 3-D homeostatic attractor



Normal: 2-D cycle · Fault: collapse · H-ASIZH: bounded 3-D attractor

Micro-Architecture + GLFI Framework

1. **Izhikevich neuron with the affine mapping Vhw = 64 (vbio + 65) — no multipliers, shift-add only.**
2. **Plant-controller homeostasis with Lyapunov-proven UUB stability — saturation is ruled out mathematically.**
3. **GLFI Engine Mark I: a multi-agent LLM generates the Cocotb testbenches; Shadow Hook injects SA0 / SA1 faults.**
4. **Four-valued logic (0,1,X,Z) models bridging and stuck-open defects that binary simulators cannot express.**
5. **An exhaustive campaign sweeps five PPA models, from the bare baseline through to full TMR.**

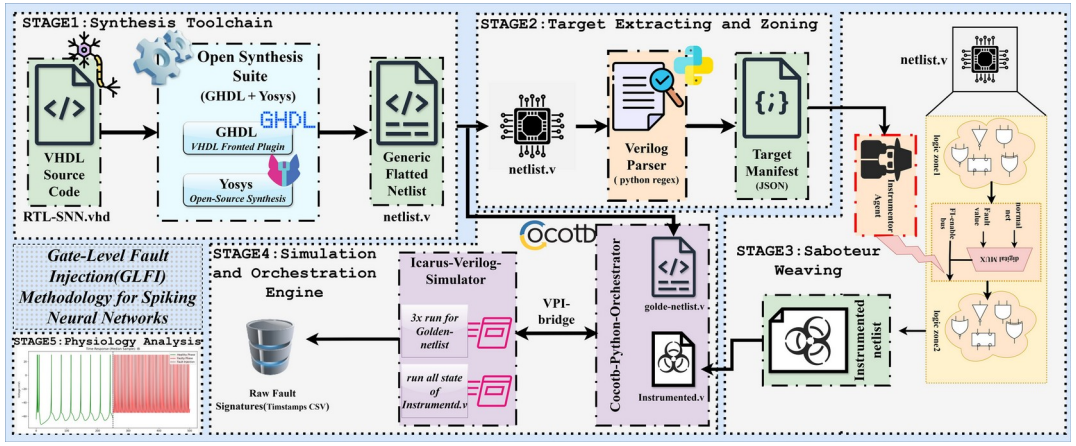
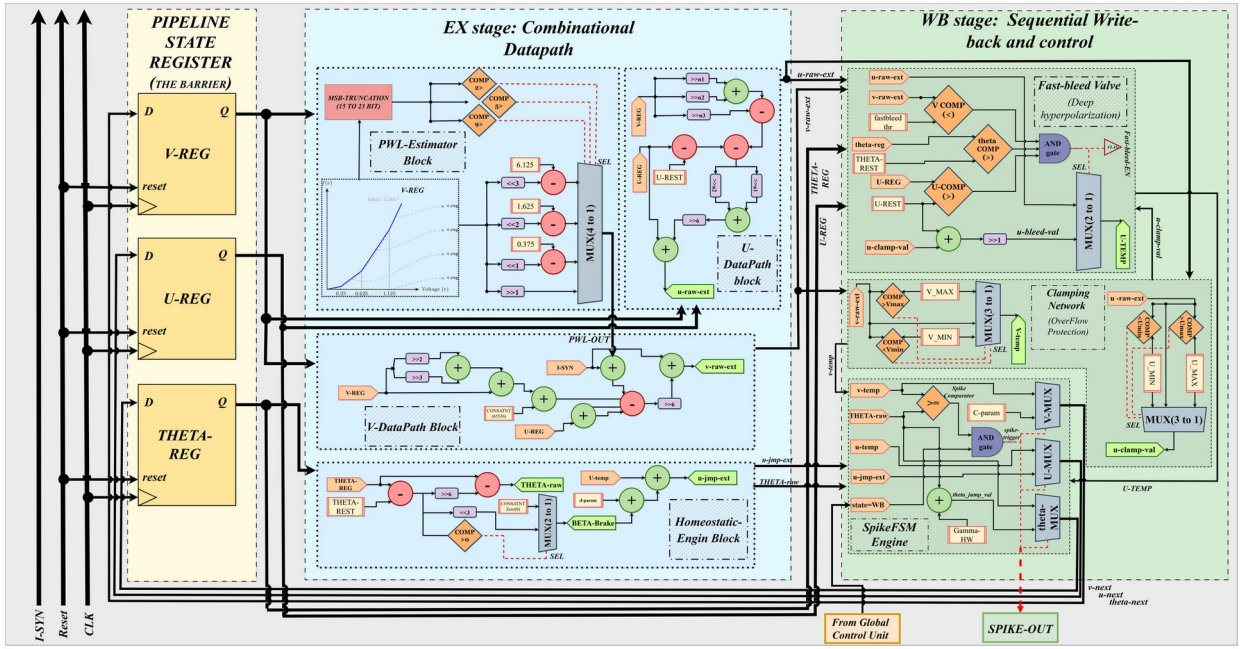


Fig1:GLFI engine — fault-injection architecture

Fig2:H-ASIZH neuron core architecture



§ 4 · RESULTS

Results & Analysis

Power, performance and area · the GLFI campaign · reliability metrics · on-chip learning

95.8%
Functional survival
under exhaustive GLFI

4093
Standard cells
SkyWater 130 nm

91.4 mW
Total power
at 62 MHz

84.3%
On-chip SDSP
learning accuracy

Synthesis Results & Exhaustive Fault Injection

1. Baseline	2. Homeostasis	3. Light TMR	4. Restoring Organ
1,772 cells	**2,469 cells**	**3,969 cells**	**4,093 cells**
83.6 MHz · 30.7 mW	64.9 MHz · 52.8 mW	64.2 MHz · 88.5 mW	62.1 MHz · 91.4 mW

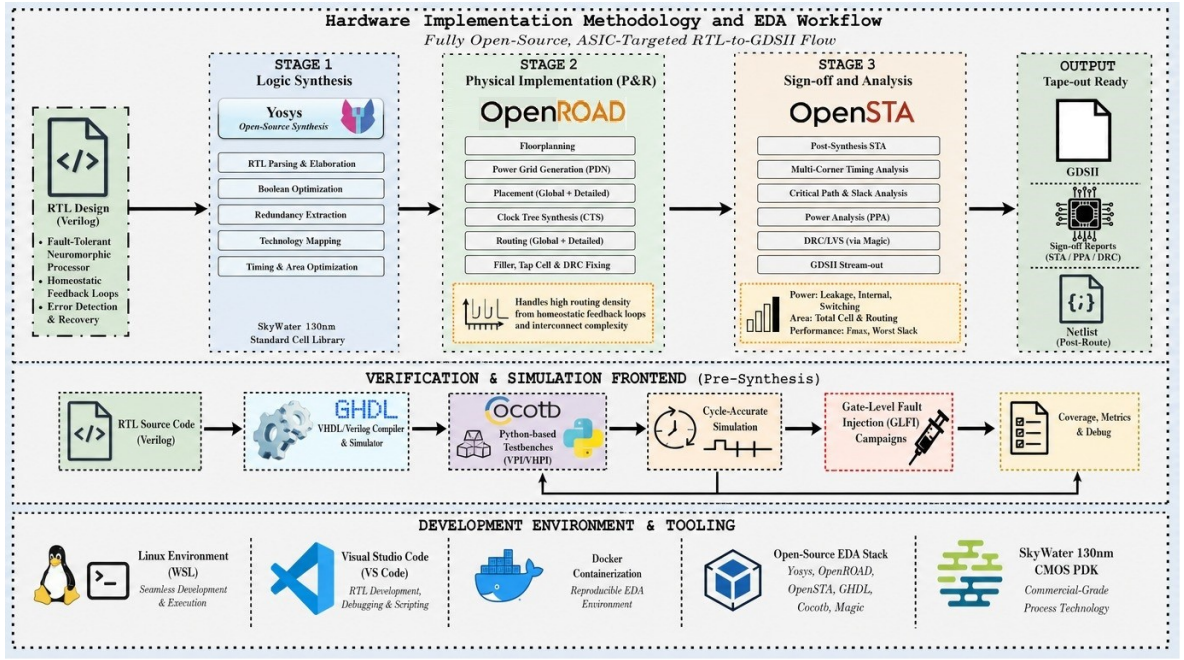
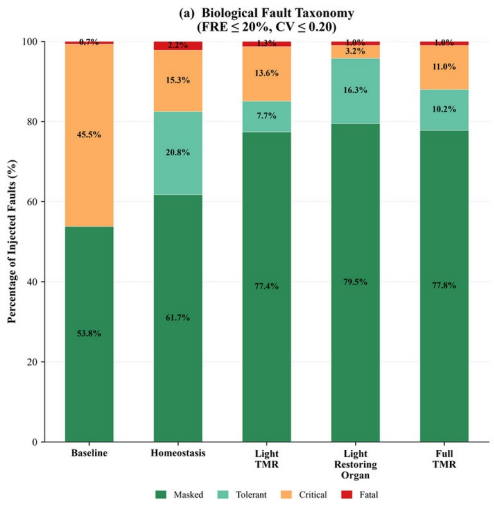


Fig1:EDA workflow — fully open-source RTL-to-GDSII flow



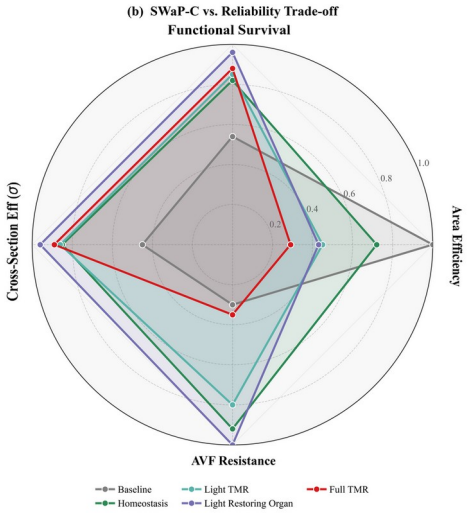


Fig2:GLFI campaign — 100% fault coverage

Four-Dimensional Reliability Analysis + On-Chip Learning

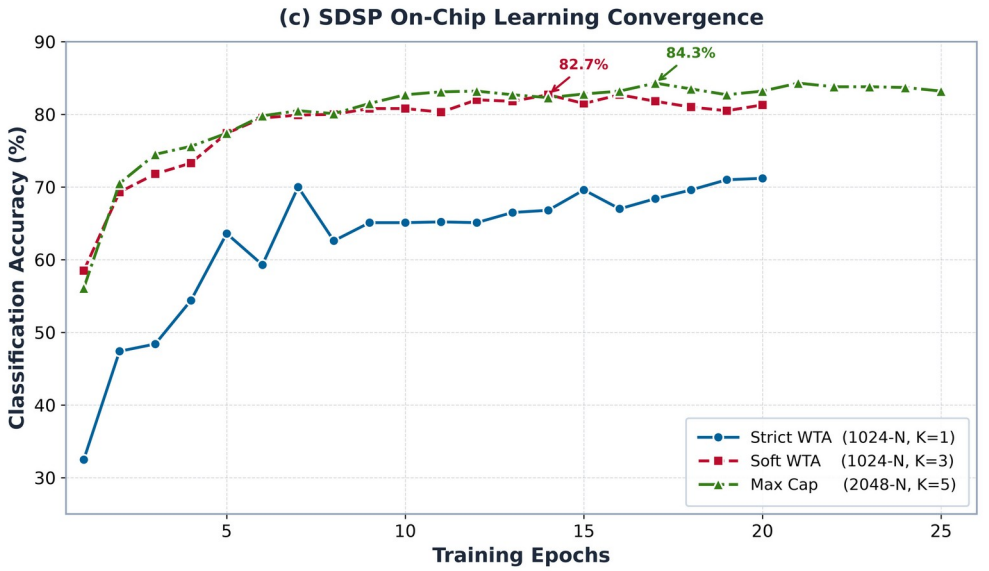
1 **Area Efficiency**
H-ASIZH adds 39% more cells, where full TMR adds 240%.

2 **Functional Survival**
The highest survival rate of all variants, through graceful degradation.

3 **AVF Resistance**
Triple defence: anti-windup clamping, fast-bleed and beta-brake.

4 **Cross-Section Efficiency (σ)**
Survival parity with full TMR at a fraction of the overhead.

SDSP on-chip learning — convergence



84.3% on-chip learning accuracy

Soft WTA (K = 3): 82.7% — converges within 15–20 epochs

From Fault Tolerance to Next-Generation Systems

1 Spiking YOLO & UAV Accelerators

R-STDP for real-time object detection, and an ASIC UAV processor fortified with adaptive homeostasis against atmospheric radiation.

2 Biocompatible Neuroprosthetics

EEG/EMG-driven on-chip learning at µW power, with thermal safety and zero-failure tolerance for bionic limbs and BCIs.

3 Spiky Transformers & LLMs

Spiking self-attention carried by spike timing — MACs replaced by accumulators, giving mW-level NLP on edge devices.

4 Self-Healing Silicon

R-STDP combined with homeostasis for online repair: healthy neighbours recalibrate and absorb the load after damage.

“Adaptive homeostasis — the DNA of tomorrow's processors.”

CLOSING STATEMENT

This work is not merely a noise-mitigation strategy — it is the DNA of tomorrow's processors.

Whether we are grafting a neuroprosthetic hand, building a radiation-hardened drone, or running an LLM on a smartwatch, we need silicon that learns, consumes µW of power, heals itself, and stays thermally optimal.

This is the road ahead for neuromorphic architectures.

Learns | µW Power | Self-Healing | Thermally Safe

H-ASIZH — Homeostatic-Adaptive Spiking Izhikevich Hardware

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